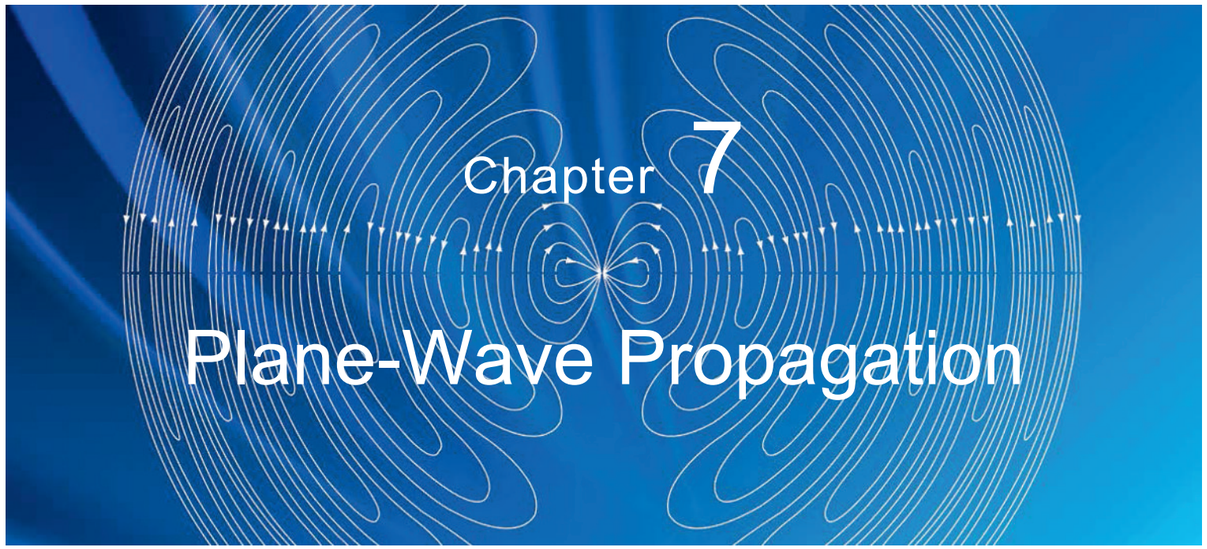


Chapter Contents

Unbounded EM Waves, 302
7-1 Time-Harmonic Fields, 303
7-2 Plane-Wave Propagation in Lossless Media, 304
7-3 Wave Polarization, 309
TB13 RFID Systems, 313
7-4 Plane-Wave Propagation in Lossy Media, 317
TB14 Liquid Crystal Display (LCD), 320
7-5 Current Flow in a Good Conductor, 325
7-6 Electromagnetic Power Density, 328
Chapter 7 Summary, 331
Problems, 333

Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Describe mathematically the electric and magnetic fields of TEM waves.
2. Describe the polarization properties of an EM wave.
3. Relate the propagation parameters of a wave to the constitutive parameters of the medium.
4. Characterize the flow of current in conductors and use it to calculate the resistance of a coaxial cable.
5. Calculate the rate of power carried by an EM wave in both lossless and lossy media.

第7章

平面波的传播

章节内容

无界电磁波、302 7-1 时谐波场、303 7-2 无损介质中的平面波传播、304 7-3 波偏振、309 TB13 RFID 系统、313 7-4 有损介质中的平面波传播、317 TB14 液晶显示器 (LCD)、320 7-5 介质中的电流良导体，325 7-6 电磁功率密度，328 第 7 章总结，331 问题，333

目标

学习本章介绍的材料后，您应该能够：

1. 用数学方法描述 TEM 波的电场和磁场。 2. 描述电磁波的偏振特性。 3. 将波的传播参数与介质的本构参数联系起来。 4. 表征导体中的电流并用它来计算同轴电缆的电阻。 5. 计算电磁波在无损和有损介质中携带的功率率。

Unbounded EM Waves

It was established in Chapter 6 that a time-varying electric field produces a magnetic field and, conversely, a time-varying magnetic field produces an electric field. This cyclic pattern often results in electromagnetic (EM) waves propagating through free space and in material media. When a wave propagates through a homogeneous medium without interacting with obstacles or material interfaces, it is said to be ***unbounded***. Light waves emitted by the sun and radio transmissions by antennas are good examples. Unbounded waves may propagate in both lossless and lossy media. Waves propagating in a ***lossless medium*** (e.g., air and perfect dielectrics) are similar to those on a lossless transmission line in that they do not attenuate. When propagating in a ***lossy medium*** (material with nonzero conductivity, such as water), part of the power carried by an EM wave gets converted into heat. A wave produced by a localized source, such as an antenna, expands outwardly in the form of a ***spherical wave***, as depicted in **Fig. 7-1(a)**. Even though an antenna may radiate more energy along some directions than along others, the spherical wave travels at the same speed in all directions. To an observer very far away from the source, however, the ***wavefront*** of the spherical wave appears approximately *planar*, as if it were part of a ***uniform plane wave*** with identical properties at all points in the plane tangent to the wavefront (**Fig. 7-1(b)**). Plane waves are easily described using a Cartesian coordinate system, which is mathematically easier to work with than the spherical coordinate system needed to describe spherical waves.

When a wave propagates along a material structure, it is said to be ***guided***. The Earth's surface and ionosphere constitute parallel boundaries of a natural structure capable of guiding short-wave radio transmissions in the HF band* (3 to 30 MHz); indeed, the ionosphere is a good reflector at these frequencies, thereby allowing the waves to zigzag between the two boundaries (**Fig. 7-2**). When we discussed wave propagation on a transmission line in Chapter 2, we dealt with voltages and currents. For a transmission-line circuit such as that shown in **Fig. 7-3**, the ac voltage source excites an incident wave that travels down the coaxial line toward the load, and unless the load is matched to the line, part (or all) of the incident wave is reflected back toward the generator. At any point on the line, the instantaneous total voltage $v(z,t)$ is the sum of the incident and reflected waves—both of which vary sinusoidally with time. Associated with the voltage difference between the inner and outer conductors of the coaxial line is a radial electric field $\mathbf{E}(z,t)$ that exists in the dielectric material between the conductors, and since $v(z,t)$ varies sinusoidally with time, so does $\mathbf{E}(z,t)$. Furthermore, the current flowing through the inner

*See **Fig. 1-17**.

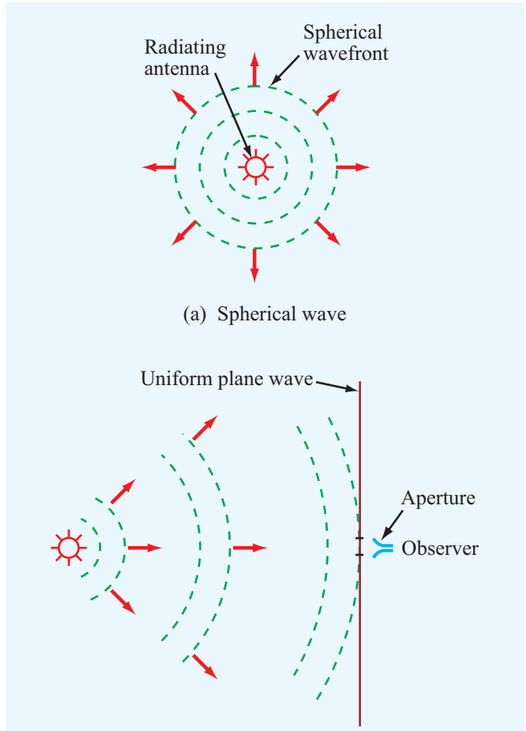


Figure 7-1 Waves radiated by an EM source, such as a light bulb or an antenna, have spherical wavefronts, as in (a); to a distant observer, however, the wavefront across the observer's aperture appears approximately planar, as in (b).

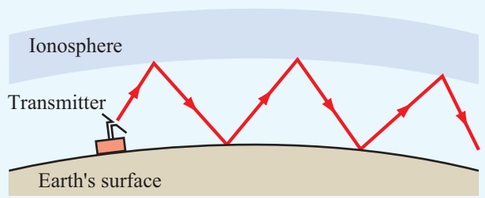


Figure 7-2 The atmospheric layer bounded by the ionosphere at the top and the Earth's surface at the bottom forms a guiding structure for the propagation of radio waves in the HF band.

无界电磁波

第六章确立了时变电场产生磁场，反之，时变磁场产生电场。这种循环模式通常会导致电磁 (EM) 波在自由空间和材料介质中传播。当波在均匀介质中传播而不与障碍物或材料界面相互作用时，称为无界波。太阳发射的光波和天线的无线电传输就是很好的例子。无界波可以在无损和有损介质中传播。在无损介质（例如空气和理想电介质）中传播的波与在无损传输线上传播的波相似，因为它们不会衰减。当在有损介质（具有非零电导率的材料，例如水）中传播时，电磁波携带的部分功率会转化为热量。由局部源（例如天线）产生的波以球面波的形式向外扩展，如图 7-1（a）所示。尽管天线沿某些方向辐射的能量可能比沿其他方向辐射的能量多，但球面波在所有方向上以相同的速度传播。然而，对于距离源很远的观察者来说，球面波的波前看起来近似平面，就好像它是均匀平面波的一部分，在与波前相切的平面上的所有点都具有相同的属性（图7-1（b））。使用笛卡尔坐标系可以轻松地描述平面波，该坐标系在数学上比描述球面波所需的球面坐标系更容易使用。

当波沿着材料结构传播时，称为被引导。地球表面和电离层构成自然结构的平行边界，能够引导 HF 频段* (3 至 30 MHz) 的短波无线电传输；事实上，电离层在这些频率下是一个很好的反射器，从而允许波在两个边界之间曲折（图7-2）。当我们在第二章中讨论传输线上的波传播时，我们讨论了电压和电流。对于如图 7-3 所示的传输线电路，交流电压源激发沿同轴线向负载传播的入射波，除非负载与线路匹配，否则部分（或全部）入射波将反射回发生器。在线路上的任何一点，瞬时总电压 v(z, t) 是入射波和反射波的总和，两者都随时间呈正弦变化。与同轴线内导体和外导体之间的电压差相关的是导体之间的介电材料中存在的径向电场 E(z,t)，并且由于 v(z,t) 随时间呈正弦变化，因此 E(z,t) 也随之变化。此外，流过内部的电流

*See **Fig. 1-17**.

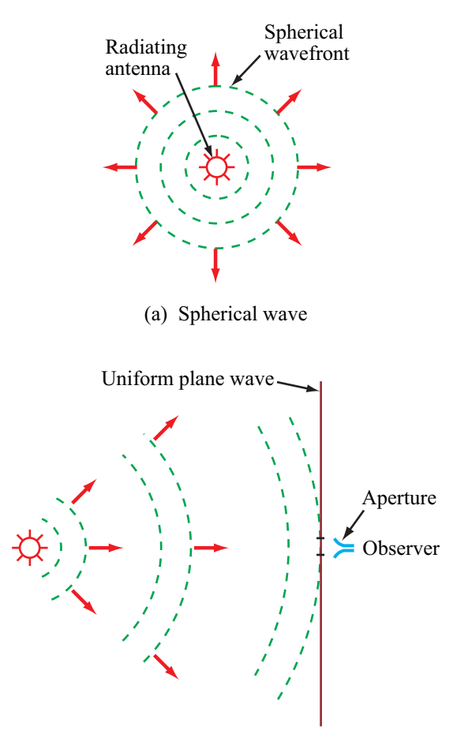


图 7-1 电磁源（例如灯泡或天线）辐射的波具有球形波前，如 (a) 所示；然而，对于远处的观察者来说，穿过观察者孔径的波前看起来近似平面，如（b）所示。

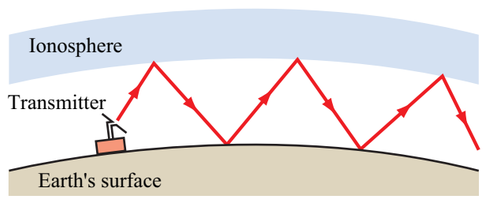


图 7-2 顶部为电离层、底部为地球表面的大气层形成了 HF 频段无线电波传播的引导结构。

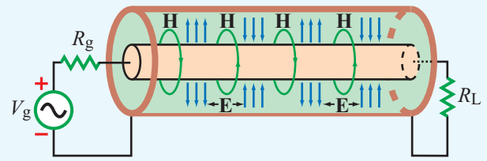


Figure 7-3 A guided electromagnetic wave traveling in a coaxial transmission line consists of time-varying electric and magnetic fields in the dielectric medium between the inner and outer conductors.

conductor induces an azimuthal magnetic field $\mathbf{H}(z,t)$ in the dielectric material surrounding it. These coupled fields, $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$, constitute an electromagnetic wave. Thus, we can model wave propagation on a transmission line either in terms of the voltages across the line and the currents in its conductors or in terms of the electric and magnetic fields in the dielectric medium between the conductors.

In this chapter, we focus our attention on wave propagation in unbounded media. Unbounded waves have many practical applications in science and engineering. We consider both lossless and lossy media. Even though strictly speaking uniform plane waves cannot exist, we study them in this chapter to develop a physical understanding of wave propagation in lossless and lossy media. In Chapter 8, we examine how both planar and spherical waves are reflected by and transmitted through boundaries between dissimilar media. The proc esses of radiation and reception of waves by antennas are treated in Chapter 9.

7-1 Time-Harmonic Fields

Time-varying electric and magnetic fields (**E**, **D**, **B**, and **H**) and their sources (the charge density ρ_v and current density **J**) generally depend on the spatial coordinates (x,y,z) and the time variable t. However, if their time variation is sinusoidal with angular frequency ω, then these quantities can be represented by a phasor that depends on (x,y,z) only. The vector phasor $\widetilde{\mathbf{E}}(x,y,z)$ and the instantaneous field $\mathbf{E}(x,y,z;t)$ it describes are related as

$$\mathbf{E}(x,y,z;t) = \Re\mathrm{e}\left[\widetilde{\mathbf{E}}(x,y,z)\, e^{j\omega t}\right]. \tag{7.1}$$

Similar definitions apply to **D**, **B**, and **H**, as well as to ρ_v and **J**. For a linear, isotropic, and homogeneous medium with electrical permittivity ε, magnetic permeability μ, and conductivity σ, Maxwell's equations (6.1) to (6.4) assume the following form in the phasor domain.

$$\nabla\cdot\widetilde{\mathbf{E}} = \tilde{\rho}_v/\varepsilon, \tag{7.2a}$$
$$\nabla\times\widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}}, \tag{7.2b}$$
$$\nabla\cdot\widetilde{\mathbf{H}} = 0, \tag{7.2c}$$
$$\nabla\times\widetilde{\mathbf{H}} = \widetilde{\mathbf{J}} + j\omega\varepsilon\widetilde{\mathbf{E}}. \tag{7.2d}$$

To derive these equations, we used $\mathbf{D} = \varepsilon\mathbf{E}$ and $\mathbf{B} = \mu\mathbf{H}$ as well as the fact that, for time-harmonic quantities, differentiation in the time domain corresponds to multiplication by $j\omega$ in the phasor domain. These equations are the starting point for the subject matter treated in this chapter.

7-1.1 Complex Permittivity

In a medium with conductivity σ, the conduction current density $\widetilde{\mathbf{J}}$ is related to $\widetilde{\mathbf{E}}$ by $\widetilde{\mathbf{J}} = \sigma\widetilde{\mathbf{E}}$. Assuming no other current flows in the medium, Eq. (7.2d) may be written as

$$\nabla\times\widetilde{\mathbf{H}} = \widetilde{\mathbf{J}} + j\omega\varepsilon\widetilde{\mathbf{E}} = (\sigma + j\omega\varepsilon)\widetilde{\mathbf{E}} = j\omega\left(\varepsilon - j\frac{\sigma}{\omega}\right)\widetilde{\mathbf{E}}. \tag{7.3}$$

By defining the ***complex permittivity*** ε_c as

$$\varepsilon_c = \varepsilon - j\frac{\sigma}{\omega}, \tag{7.4}$$

Eq. (7.3) can be rewritten as

$$\nabla\times\widetilde{\mathbf{H}} = j\omega\varepsilon_c\widetilde{\mathbf{E}}. \tag{7.5}$$

In a source-free medium, $\rho_v = 0$. Hence, Maxwell's equations become

$$\nabla\cdot\widetilde{\mathbf{E}} = 0, \tag{7.6a}$$
$$\nabla\times\widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}}, \tag{7.6b}$$
$$\nabla\cdot\widetilde{\mathbf{H}} = 0, \tag{7.6c}$$
$$\nabla\times\widetilde{\mathbf{H}} = j\omega\varepsilon_c\widetilde{\mathbf{E}}. \tag{7.6d}$$

The complex permittivity ε_c given by Eq. (7.4) is often written in terms of a real part ε' and an imaginary part ε''. Thus,

$$\varepsilon_c = \varepsilon - j\frac{\sigma}{\omega} = \varepsilon' - j\varepsilon'', \tag{7.7}$$

with

$$\varepsilon' = \varepsilon, \tag{7.8a}$$
$$\varepsilon'' = \frac{\sigma}{\omega}. \tag{7.8b}$$

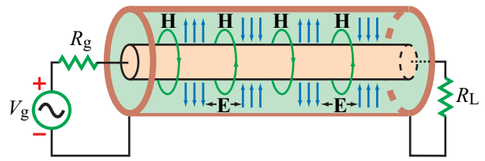


图 7-3 在同轴传输线中传播的引导电磁波由内导体和外导体之间的介电介质中的时变电场和磁场组成。

导体在其周围的介电材料中感应出方位角磁场 H(z,t)。这些耦合场 E(z,t) 和 H(z,t) 构成电磁波。因此，我们可以根据传输线上的电压和导体中的电流，或者根据导体之间介电介质中的电场和磁场来模拟传输线上的波传播。

在本章中，我们将重点关注无界介质中的波传播。无界波在科学和工程中有许多实际应用。我们考虑无损和有损媒体。尽管严格来说均匀平面波不存在，但我们在本章中研究它们是为了对波在无损和有损介质中传播的物理理解。在第八章中，我们研究平面波和球面波如何在不同介质之间的边界反射和传播。第 9 章讨论天线辐射和接收电波的过程。

7-1 时谐波场

时变电场和磁场（E、D、B 和 H）及其源（电荷密度 ρ_v 和电流密度 J）通常取决于空间坐标（x、y、z）和时间变量 t。然而，如果它们的时间变化是具有角频率 ω 的正弦曲线，则这些量可以用仅依赖于 (x, y, z) 的相量来表示。矢量相量 ~E(x, y, z) 和它描述的瞬时场 E(x, y, z;t) 的关系为

$$\mathbf{E}(x,y,z;t) = \Re\mathrm{e}\left[\widetilde{\mathbf{E}}(x,y,z)\, e^{j\omega t}\right]. \tag{7.1}$$

类似的定义适用于 D、B 和 H，以及 ρ_v 和 J。对于具有介电常数 ε、磁导率 μ 和电导率 σ 的线性、各向同性和均匀介质，麦克斯韦方程 (6.1) 至 (6.4) 假设相量域中的以下形式。

$$\nabla\cdot\widetilde{\mathbf{E}} = \tilde{\rho}_v/\varepsilon, \tag{7.2a}$$
$$\nabla\times\widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}}, \tag{7.2b}$$
$$\nabla\cdot\widetilde{\mathbf{H}} = 0, \tag{7.2c}$$
$$\nabla\times\widetilde{\mathbf{H}} = \widetilde{\mathbf{J}} + j\omega\varepsilon\widetilde{\mathbf{E}}. \tag{7.2d}$$

为了推导这些方程，我们使用了 D = ε E 和 B = μ H 以及以下事实：对于时谐量，时域中的微分对应于相量域中 j ω 的乘法。这些方程是本章讨论的主题的起点。

7-1.1 复介电常数

在电导率为σ的介质中，传导电流密度~J与~E的关系为~J = σ E。假设介质中没有其他电流流动，方程：(7.2d) 可以写成

$$\nabla\times\widetilde{\mathbf{H}} = \widetilde{\mathbf{J}} + j\omega\varepsilon\widetilde{\mathbf{E}} = (\sigma + j\omega\varepsilon)\widetilde{\mathbf{E}} = j\omega\left(\varepsilon - j\frac{\sigma}{\omega}\right)\widetilde{\mathbf{E}}. \tag{7.3}$$

将复介电常数 ε_c 定义为

$$\varepsilon_c = \varepsilon - j\frac{\sigma}{\omega}, \tag{7.4}$$

等式。(7.3) 可以重写为

$$\nabla\times\widetilde{\mathbf{H}} = j\omega\varepsilon_c\widetilde{\mathbf{E}}. \tag{7.5}$$

在无源介质中，$\rho_v = 0$。因此，麦克斯韦方程组变为

$$\nabla\cdot\widetilde{\mathbf{E}} = 0, \tag{7.6a}$$
$$\nabla\times\widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}}, \tag{7.6b}$$
$$\nabla\cdot\widetilde{\mathbf{H}} = 0, \tag{7.6c}$$
$$\nabla\times\widetilde{\mathbf{H}} = j\omega\varepsilon_c\widetilde{\mathbf{E}}. \tag{7.6d}$$

复介电常数 ε_c 由方程式给出。(7.4) 通常用实部 ε' 和虚部 ε'' 来表示。因此，

$$\varepsilon_c = \varepsilon - j\frac{\sigma}{\omega} = \varepsilon' - j\varepsilon'', \tag{7.7}$$

和

$$\varepsilon' = \varepsilon, \tag{7.8a}$$
$$\varepsilon'' = \frac{\sigma}{\omega}. \tag{7.8b}$$

For a lossless medium with $\sigma = 0$, it follows that $\varepsilon'' = 0$ and $\varepsilon_c = \varepsilon' = \varepsilon$.

7-1.2 Wave Equations

Next, we derive wave equations for $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ and then solve them to obtain explicit expressions for $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ as a function of the spatial variables (x, y, z). To that end, we start by taking the curl of both sides of Eq. (7.6b) to get

$$\nabla \times (\nabla \times \widetilde{\mathbf{E}}) = -j\omega\mu(\nabla \times \widetilde{\mathbf{H}}). \tag{7.9}$$

Upon substituting Eq. (7.6d) into Eq. (7.9), we obtain

$$\nabla \times (\nabla \times \widetilde{\mathbf{E}}) = -j\omega\mu(j\omega\varepsilon_c\widetilde{\mathbf{E}}) = \omega^2\mu\varepsilon_c\widetilde{\mathbf{E}}. \tag{7.10}$$

From Eq. (3.113), we know that the curl of the curl of $\widetilde{\mathbf{E}}$ is

$$\nabla \times (\nabla \times \widetilde{\mathbf{E}}) = \nabla(\nabla \cdot \widetilde{\mathbf{E}}) - \nabla^2\widetilde{\mathbf{E}}, \tag{7.11}$$

where $\nabla^2\widetilde{\mathbf{E}}$ is the Laplacian of $\widetilde{\mathbf{E}}$, which in Cartesian coordinates is given by

$$\nabla^2\widetilde{\mathbf{E}} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\widetilde{\mathbf{E}}. \tag{7.12}$$

In view of Eq. (7.6a), the use of Eq. (7.11) in Eq. (7.10) gives

$$\nabla^2\widetilde{\mathbf{E}} + \omega^2\mu\varepsilon_c\widetilde{\mathbf{E}} = 0, \tag{7.13}$$

which is known as the ***homogeneous wave equation for*** $\widetilde{\mathbf{E}}$. By defining the ***propagation constant*** γ as

$$\gamma^2 = -\omega^2\mu\varepsilon_c, \tag{7.14}$$

Eq. (7.13) can be written as

$$\nabla^2\widetilde{\mathbf{E}} - \gamma^2\widetilde{\mathbf{E}} = 0. \quad \textbf{(wave equation for } \widetilde{\mathbf{E}}\textbf{)} \tag{7.15}$$

To derive Eq. (7.15), we took the curl of both sides of Eq. (7.6b) and then we used Eq. (7.6d) to eliminate $\widetilde{\mathbf{H}}$ and obtain an equation in $\widetilde{\mathbf{E}}$ only. If we reverse the process, that is, if we start by taking the curl of both sides of Eq. (7.6d) and then use Eq. (7.6b) to eliminate $\widetilde{\mathbf{E}}$, we obtain a wave equation for $\widetilde{\mathbf{H}}$:

$$\nabla^2\widetilde{\mathbf{H}} - \gamma^2\widetilde{\mathbf{H}} = 0. \quad \textbf{(wave equation for } \widetilde{\mathbf{H}}\textbf{)} \tag{7.16}$$

Since the wave equations for $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ are of the same form, so are their solutions.

7-2 Plane-Wave Propagation in Lossless Media

The properties of an electromagnetic wave, such as its phase velocity u_p and wavelength λ, depend on the angular frequency ω and the medium's three constitutive parameters: ε, μ, and σ. If the medium is ***nonconducting*** ($\sigma = 0$), the wave does not suffer any attenuation as it travels; hence, the medium is said to be ***lossless***. Because in a lossless medium $\varepsilon_c = \varepsilon$, Eq. (7.14) becomes

$$\gamma^2 = -\omega^2\mu\varepsilon. \tag{7.17}$$

For lossless media, it is customary to define the ***wavenumber*** k as

$$k = \omega\sqrt{\mu\varepsilon}. \tag{7.18}$$

In view of Eq. (7.17), $\gamma^2 = -k^2$ and Eq. (7.15) becomes

$$\nabla^2\widetilde{\mathbf{E}} + k^2\widetilde{\mathbf{E}} = 0. \tag{7.19}$$

7-2.1 Uniform Plane Waves

For an electric field phasor defined in Cartesian coordinates as

$$\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\widetilde{E}_x + \hat{\mathbf{y}}\widetilde{E}_y + \hat{\mathbf{z}}\widetilde{E}_z, \tag{7.20}$$

substitution of Eq. (7.12) into Eq. (7.19) gives

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)(\hat{\mathbf{x}}\widetilde{E}_x + \hat{\mathbf{y}}\widetilde{E}_y + \hat{\mathbf{z}}\widetilde{E}_z) + k^2(\hat{\mathbf{x}}\widetilde{E}_x + \hat{\mathbf{y}}\widetilde{E}_y + \hat{\mathbf{z}}\widetilde{E}_z) = 0. \tag{7.21}$$

To satisfy Eq. (7.21), each vector component on the left-hand side of the equation must vanish. Hence,

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2\right)\widetilde{E}_x = 0, \tag{7.22}$$

and similar expressions apply to $\widetilde{E}_y$ and $\widetilde{E}_z$.

► A ***uniform plane wave*** is characterized by electric and magnetic fields that have uniform properties at all points across an infinite plane. ◄

If this happens to be the x–y plane, then $\mathbf{E}$ and $\mathbf{H}$ do not vary with x or y. Hence, $\partial\widetilde{E}_x/\partial x = 0$ and $\partial\widetilde{E}_x/\partial y = 0$, so Eq. (7.22) reduces to

$$\frac{d^2\widetilde{E}_x}{dz^2} + k^2\widetilde{E}_x = 0. \tag{7.23}$$

304 CHAPTER 7 PLANE-WAVE PROPAGATION

对于 σ = 0 的无损介质，可以得出 ε″ = 0 和 $\varepsilon_c = \varepsilon' = \varepsilon$ 。

7-1.2波动方程

接下来，我们推导 ~E 和 ~H 的波动方程，然后求解它们以获得 ~E 和 ~H 作为空间变量 (x, y, z) 的函数的显式表达式。为此，我们首先取方程两边的旋度。 (7.6b) 得到

$$\nabla \times (\nabla \times \widetilde{\mathbf{E}}) = -j\omega\mu(\nabla \times \widetilde{\mathbf{H}}). \tag{7.9}$$

代入方程式后。 (7.6d) 代入方程。 (7.9) ，我们得到

$$\nabla \times (\nabla \times \widetilde{\mathbf{E}}) = -j\omega\mu(j\omega\varepsilon_c\widetilde{\mathbf{E}}) = \omega^2\mu\varepsilon_c\widetilde{\mathbf{E}}. \tag{7.10}$$

从方程。 (3.113) ，我们知道~E的旋度的旋度是

$$\nabla \times (\nabla \times \widetilde{\mathbf{E}}) = \nabla(\nabla \cdot \widetilde{\mathbf{E}}) - \nabla^2\widetilde{\mathbf{E}}, \tag{7.11}$$

其中 ∇^2~E 是 ~E 的拉普拉斯算子，在笛卡尔坐标中由下式给出

$$\nabla^2\widetilde{\mathbf{E}} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\widetilde{\mathbf{E}}. \tag{7.12}$$

鉴于方程。 (7.6a) ，使用方程。 (7.11) 式(7.10) 给出

$$\nabla^2\widetilde{\mathbf{E}} + \omega^2\mu\varepsilon_c\widetilde{\mathbf{E}} = 0, \tag{7.13}$$

这被称为 ~E 的齐次波动方程。通过将传播常数 γ 定义为

$$\gamma^2 = -\omega^2\mu\varepsilon_c, \tag{7.14}$$

等式。 (7.13) 可以写成

$$\nabla^2\widetilde{\mathbf{E}} - \gamma^2\widetilde{\mathbf{E}} = 0. \quad \textbf{(wave equation for } \widetilde{\mathbf{E}}\textbf{)} \tag{7.15}$$

推导出方程。 (7.15)，我们取方程两边的旋度。 (7.6b) 然后我们使用方程。 (7.6d) 消去~H 并仅得到~E 中的方程。如果我们反转这个过程，也就是说，如果我们从方程两边的旋度开始。 (7.6d) ，然后使用方程。 (7.6b) 消除~E，我们得到~H的波动方程：

$$\nabla^2\widetilde{\mathbf{H}} - \gamma^2\widetilde{\mathbf{H}} = 0. \quad \textbf{(wave equation for } \widetilde{\mathbf{H}}\textbf{)} \tag{7.16}$$

由于 ~E 和 ~H 的波动方程具有相同的形式，因此它们的解也具有相同的形式。

7-2 无损介质中的平面波传播

电磁波的特性，例如相速度 u_p 和波长 λ，取决于角频率 ω 和介质的三个本构参数：ε、μ 和 σ。如果介质不导电 (σ = 0)，则波在传播时不会受到任何衰减；因此，该介质被称为无损介质。因为在无损介质 $\varepsilon_c = \varepsilon$ 中，等式： (7.14) 变为

$$\gamma^2 = -\omega^2\mu\varepsilon. \tag{7.17}$$

对于无损介质，通常将波数 k 定义为

$$k = \omega\sqrt{\mu\varepsilon}. \tag{7.18}$$

鉴于方程。 (7.17), $\gamma^2 = -k^2$ 和方程。 (7.15) 变为

$$\nabla^2\widetilde{\mathbf{E}} + k^2\widetilde{\mathbf{E}} = 0. \tag{7.19}$$

7-2.1 均匀平面波

对于笛卡尔坐标中定义的电场相量为

$$\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\widetilde{E}_x + \hat{\mathbf{y}}\widetilde{E}_y + \hat{\mathbf{z}}\widetilde{E}_z, \tag{7.20}$$

代入方程式（7.12）代入等式。 (7.19) 给出

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)(\hat{\mathbf{x}}\widetilde{E}_x + \hat{\mathbf{y}}\widetilde{E}_y + \hat{\mathbf{z}}\widetilde{E}_z) + k^2(\hat{\mathbf{x}}\widetilde{E}_x + \hat{\mathbf{y}}\widetilde{E}_y + \hat{\mathbf{z}}\widetilde{E}_z) = 0. \tag{7.21}$$

满足方程。根据 (7.21) 式，方程左侧的每个向量分量都必须消失。因此，

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2\right)\widetilde{E}_x = 0, \tag{7.22}$$

类似的表达式适用于 ~E_y 和 ~E_z。

► 均匀平面波的特征是电场和磁场在无限平面上的所有点上具有均匀的特性。 ◄

如果这恰好是 x-y 平面，则 E 和 H 不随 x 或 y 变化。因此，∂~$E_x/\partial x = 0$ 和 ∂~$E_x/\partial y = 0$，所以等式： (7.22) 减少为

$$\frac{d^2\widetilde{E}_x}{dz^2} + k^2\widetilde{E}_x = 0. \tag{7.23}$$

Similar expressions apply to $\widetilde{E}_y$, $\widetilde{H}_x$, and $\widetilde{H}_y$. The remaining components of $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ are zero; that is, $\widetilde{E}_z = \widetilde{H}_z = 0$. To show that $\widetilde{E}_z = 0$, let us consider the z component of Eq. (7.6d),

$$\hat{\mathbf{z}}\left(\frac{\partial \widetilde{H}_y}{\partial x} - \frac{\partial \widetilde{H}_x}{\partial y}\right) = \hat{\mathbf{z}}\, j\omega\varepsilon \widetilde{E}_z. \tag{7.24}$$

Since $\partial \widetilde{H}_y/\partial x = \partial \widetilde{H}_x/\partial y = 0$, it follows that $\widetilde{E}_z = 0$. A similar examination involving Eq. (7.6b) reveals that $\widetilde{H}_z = 0$.

► This means that a plane wave has no electric-field or magnetic-field components along its direction of propagation. ◄

For the phasor quantity $\widetilde{E}_x$, the general solution of the ordinary differential equation given by Eq. (7.23) is

$$\widetilde{E}_x(z) = \widetilde{E}_x^+(z) + \widetilde{E}_x^-(z) = E_{x0}^+ e^{-jkz} + E_{x0}^- e^{jkz}, \tag{7.25}$$

where E_{x0}^+ and E_{x0}^- are constants to be determined from boundary conditions. The solution given by Eq. (7.25) is similar in form to the solution for the phasor voltage $\widetilde{V}(z)$ given by Eq. (2.54a) for the lossless transmission line. The first term in Eq. (7.25), containing the negative exponential e^{-jkz}, represents a wave with amplitude E_{x0}^+ traveling in the $+z$ direction. Likewise, the second term (with e^{jkz}) represents a wave with amplitude E_{x0}^- traveling in the $-z$ direction. Assume for the time being that $\widetilde{\mathbf{E}}$ only has a component along x (i.e., $\widetilde{E}_y = 0$) and that $\widetilde{E}_x$ is associated with a wave traveling in the $+z$ direction only (i.e., $E_{x0}^- = 0$). Under these conditions,

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\widetilde{E}_x^+(z) = \hat{\mathbf{x}}E_{x0}^+ e^{-jkz}. \tag{7.26}$$

To find the magnetic field $\widetilde{\mathbf{H}}$ associated with this wave, we apply Eq. (7.6b) with $\widetilde{E}_y = \widetilde{E}_z = 0$:

$$\nabla \times \widetilde{\mathbf{E}} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ \widetilde{E}_x^+(z) & 0 & 0 \end{vmatrix} = -j\omega\mu(\hat{\mathbf{x}}\widetilde{H}_x + \hat{\mathbf{y}}\widetilde{H}_y + \hat{\mathbf{z}}\widetilde{H}_z). \tag{7.27}$$

For a uniform plane wave traveling in the $+z$ direction,

$$\partial E_x^+(z)/\partial x = \partial E_x^+(z)/\partial y = 0.$$

Hence, Eq. (7.27) gives

$$\widetilde{H}_x = 0, \tag{7.28a}$$

$$\widetilde{H}_y = \frac{1}{-j\omega\mu}\frac{\partial \widetilde{E}_x^+(z)}{\partial z}, \tag{7.28b}$$

$$\widetilde{H}_z = \frac{1}{-j\omega\mu}\frac{\partial E_x^+(z)}{\partial y} = 0. \tag{7.28c}$$

Use of Eq. (7.26) in Eq. (7.28b) gives

$$\widetilde{H}_y(z) = \frac{k}{\omega\mu}E_{x0}^+ e^{-jkz} = H_{y0}^+ e^{-jkz}, \tag{7.29}$$

where H_{y0}^+ is the amplitude of $\widetilde{H}_y(z)$ and is given by

$$H_{y0}^+ = \frac{k}{\omega\mu}E_{x0}^+. \tag{7.30}$$

For a wave traveling from the source toward the load on a transmission line, the amplitudes of its voltage and current phasors, V_0^+ and I_0^+, are related by the characteristic impedance of the line, Z_0. A similar connection exists between the electric and magnetic fields of an electromagnetic wave. The ***intrinsic impedance*** of a lossless medium is defined as

$$\eta = \frac{\omega\mu}{k} = \frac{\omega\mu}{\omega\sqrt{\mu\varepsilon}} = \sqrt{\frac{\mu}{\varepsilon}} \quad (\Omega), \tag{7.31}$$

where we used the expression for k given by Eq. (7.18).

In view of Eq. (7.31), the electric and magnetic fields of a $+z$-propagating plane wave with $\mathbf{E}$ field along $\hat{\mathbf{x}}$ are

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\widetilde{E}_x^+(z) = \hat{\mathbf{x}}E_{x0}^+ e^{-jkz}, \tag{7.32a}$$

$$\widetilde{\mathbf{H}}(z) = \hat{\mathbf{y}}\frac{\widetilde{E}_x^+(z)}{\eta} = \hat{\mathbf{y}}\frac{E_{x0}^+}{\eta}e^{-jkz}. \tag{7.32b}$$

► The electric and magnetic fields of a plane wave are perpendicular to each other, and both are perpendicular to the direction of wave travel (**Fig. 7-4**). These attributes qualify the wave as ***transverse electromagnetic*** (***TEM***). ◄

Other examples of TEM waves include waves traveling on coaxial transmission lines ($\mathbf{E}$ is along $\hat{\mathbf{r}}$, $\mathbf{H}$ is along $\hat{\boldsymbol{\phi}}$, and the direction of travel is along $\hat{\mathbf{z}}$) and spherical waves radiated by antennas.

In the general case, E_{x0}^+ is a complex quantity with magnitude $|E_{x0}^+|$ and phase angle ϕ^+. That is,

$$E_{x0}^+ = |E_{x0}^+|e^{j\phi^+}. \tag{7.33}$$

The instantaneous electric and magnetic fields therefore are

$$\mathbf{E}(z,t) = \Re\mathfrak{e}\left[\widetilde{\mathbf{E}}(z)\, e^{j\omega t}\right] = \hat{\mathbf{x}}|E_{x0}^+|\cos(\omega t - kz + \phi^+) \quad (\text{V/m}), \tag{7.34a}$$

类似的表达式适用于 ˜Ey、˜Hx 和 ˜Hy。˜E和˜H的其余分量为零；即，˜Ez = ˜Hz = 0。为了证明 ˜Ez = 0，让我们考虑方程的 z 分量。（7.6天），

$$\hat{\mathbf{z}}\left(\frac{\partial \widetilde{H}_y}{\partial x} - \frac{\partial \widetilde{H}_x}{\partial y}\right) = \hat{\mathbf{z}}\, j\omega\varepsilon \widetilde{E}_z. \tag{7.24}$$

由于 ∂ ˜Hy/∂ x = ∂ ˜Hx/∂ y = 0，因此 ˜Ez = 0。涉及方程的类似检查。(7.6b)表明˜Hz = 0。

► 这意味着平面波沿其传播方向没有电场或磁场分量。◄

对于相量量 ˜Ex，常微分方程的通解由式（1）给出。(7.23) 是

$$\widetilde{E}_x(z) = \widetilde{E}_x^+(z) + \widetilde{E}_x^-(z) = E_{x0}^+ e^{-jkz} + E_{x0}^- e^{jkz}, \tag{7.25}$$

其中 E_{x0}^+ 和 E_{x0}^- 是根据边界条件确定的常数。方程给出的解。(7.25) 的形式与方程 (7.25) 给出的相量电压 ˜V (z) 的解类似。（2.54a）为无损传输线。等式中的第一项。式（7.25）包含负指数 e^{-jkz}，表示沿 +z 方向传播的振幅为 E_{x0}^+ 的波。同样，第二项（e^{jkz}）表示沿 −z 方向传播的振幅为 E_{x0}^- 的波。暂时假设 ˜E 仅具有沿 x 的分量（即 ˜Ey = 0），并且 ˜Ex 与仅沿 +z 方向传播的波相关联（即 $E_{x0}^- = 0$）。在这些条件下，

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\widetilde{E}_x^+(z) = \hat{\mathbf{x}}E_{x0}^+ e^{-jkz}. \tag{7.26}$$

为了找到与该波相关的磁场 ˜H，我们应用方程：(7.6b) 与 ˜Ey = ˜Ez = 0:

$$\nabla \times \widetilde{\mathbf{E}} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial & \partial & \partial \\ \partial x & \partial y & \partial z \\ \widetilde{E}_x^+(z) & 0 & 0 \end{vmatrix} = -j\omega\mu(\hat{\mathbf{x}}\widetilde{H}_x + \hat{\mathbf{y}}\widetilde{H}_y + \hat{\mathbf{z}}\widetilde{H}_z). \tag{7.27}$$

对于沿 +z 方向传播的均匀平面波，

$$\partial E_x^+(z)/\partial x = \partial E_x^+(z)/\partial y = 0.$$

因此，等式。(7.27) 给出

$$\widetilde{H}_x = 0, \tag{7.28a}$$

$$\widetilde{H}_y = \frac{1}{-j\omega\mu}\frac{\partial \widetilde{E}_x^+(z)}{\partial z}, \tag{7.28b}$$

$$\widetilde{H}_z = \frac{1}{-j\omega\mu}\frac{\partial E_x^+(z)}{\partial y} = 0. \tag{7.28c}$$

使用等式。(7.26) 式(7.28b) 给出

$$\widetilde{H}_y(z) = \frac{k}{\omega\mu}E_{x0}^+ e^{-jkz} = H_{y0}^+ e^{-jkz}, \tag{7.29}$$

其中 H_{y0}^+ 是 ˜Hy(z) 的幅度，由下式给出

$$H_{y0}^+ = \frac{k}{\omega\mu}E_{x0}^+. \tag{7.30}$$

对于传输线上从源向负载传播的波，其电压和电流相量 V_0^+ 和 I_0^+ 的幅度与线路的特征阻抗 Z_0 相关。电磁波的电场和磁场之间也存在类似的联系。无损介质的固有阻抗定义为

$$\eta = \frac{\omega\mu}{k} = \frac{\omega\mu}{\omega\sqrt{\mu\varepsilon}} = \sqrt{\frac{\mu}{\varepsilon}} \quad (\Omega), \tag{7.31}$$

其中我们使用了等式给出的 k 表达式。（7.18）。

鉴于方程。(7.31) 中，E 场沿 x̂ 传播的 +z 平面波的电场和磁场为

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\widetilde{E}_x^+(z) = \hat{\mathbf{x}}E_{x0}^+ e^{-jkz}, \tag{7.32a}$$

$$\widetilde{\mathbf{H}}(z) = \hat{\mathbf{y}}\frac{\widetilde{E}_x^+(z)}{\eta} = \hat{\mathbf{y}}\frac{E_{x0}^+}{\eta}e^{-jkz}. \tag{7.32b}$$

► 平面波的电场和磁场相互垂直，并且都垂直于波的传播方向（图7-4）。这些属性将波定义为横向电磁波 (TEM)。◄

TEM 波的其他示例包括在同轴传输线上传播的波（E 沿 r̂，H 沿 $\hat{\boldsymbol{\phi}}$，传播方向沿 ẑ）以及由天线辐射的球面波。

一般情况下，E_{x0}^+ 是一个复量，其大小为 $|E_{x0}^+|$ 和相位角 ϕ^+。那是，

$$E_{x0}^+ = |E_{x0}^+|e^{j\phi^+}. \tag{7.33}$$

因此瞬时电场和磁场为

$$\mathbf{E}(z,t) = \Re\mathfrak{e}\left[\widetilde{\mathbf{E}}(z)\, e^{j\omega t}\right] = \hat{\mathbf{x}}|E_{x0}^+|\cos(\omega t - kz + \phi^+) \quad (\text{V/m}), \tag{7.34a}$$